

Aditya Vasu Rana

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Education

Vellore Institute of Technology, Chennai

Aug 2023 – May 2027

Bachelor of Technology in Computer Science with specialization in Data Science, **CGPA : 9.45 (Branch Rank 3)**

- **Coursework:** Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Machine Learning, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence

Technical Skills

- **Programming Languages:** Python, R, C, C++ ,Java, SQL
- **Core AI/ML:** Machine Learning, Neural Networks, Transformers, Deep Learning, NLP, Computer Vision
- **Tools & Platforms:** Linux, Git, AWS, Docker, Jupyter, Streamlit , Hex
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, HuggingFace
- **Databases & Visualization:** PostgreSQL, PL/SQL, Tableau, Power BI, Data Storytelling
- **Web Technologies:** HTML, CSS, JavaScript, ReactJS, Flask, FastAPI, Tailwind, Bootstrap

Projects

HireSense AI – Intelligent Applicant Tracking System - FastAPI + React.js + PostgreSQL + NLP [↗](#)

- Built a **full-stack AI-powered recruitment platform** with JWT authentication and role-based workflows, enabling recruiters to manage job postings and candidates to apply, track applications, receive hiring updates.
- Designed an intelligent **resume-job matching pipeline** using semantic analysis to automate candidate evaluation, generate **ATS scores**, identify missing competencies, and rank applicants against job requirements.
- Implemented application tracking, candidate shortlisting/rejection workflows, recruiter feedback, interview question generation, and **AI-generated personalized learning roadmaps** to streamline the hiring process.

PDF Question Answering System - RAG + FastAPI + React.js [↗](#)

- Developed an end-to-end **Retrieval-Augmented Generation (RAG)** system integrating **Sentence Transformers**, **FAISS**, and **Google Gemini** for semantic question answering over PDF documents.
- Built a document processing pipeline with **PyMuPDF**, intelligent text chunking, embedding generation, and vector indexing to enable efficient semantic retrieval from large documents.
- Implemented a full-stack application using **FastAPI** and **React**, featuring prompt engineering, retrieval confidence scoring, and an interactive AI-powered chat interface.

Fashion Recommender System - CNN + ResNet50 + KNN + FastAPI Backend [↗](#)

- Built a **content-based fashion recommendation system** using **ResNet50 (ImageNet pretrained)** as a feature extractor to generate **2048-D deep embeddings** for 44000+ fashion product images.
- Implemented an efficient image retrieval pipeline using **GlobalMaxPooling**, **L2 normalization**, and **K-Nearest Neighbors (KNN)** to recommend the **Top-5 visually similar items**.
- Developed an interactive **Python FastAPI web application** that allows users to upload a fashion image and receive real-time recommendations using precomputed embeddings stored in `embeddings.pkl`.

Experience

Data Science and Analytics (D2C Insights) Intern, Kerala Ayurveda Ltd.

Oct 2025 – Jan 2026

- Analyzed D2C sales funnel across multiple product categories using Python and SQL, identifying key drop-offs such as low Add-to-Cart (ATC) and Cart-to-Checkout (C2C) rates for sales improvement initiatives.
- Conducted customer behavior analysis including low ATC diagnosis, abandoned cart AOV analysis, and first-product to next-likely-product mapping, generating actionable insights to optimize conversion journeys.
- Collaborated closely with the product management team to frame hypotheses, validate marketing strategies, and support data-driven decisions aimed at improving sales performance and customer engagement.

Research Experience – Mango Variety Classification using Swin Transformer

2026

- Conducted research on fine-grained mango variety classification using a Swin Transformer, integrating preprocessing, data augmentation, weighted loss, and transfer learning.
- Achieved 98.04% test accuracy across 15 classes with comprehensive evaluation using precision, recall, F1-score, and confusion matrix, documenting the methodology and experimental analysis.

Certifications and Achievements

- Frontier Research Paper on Mango Variety Classification using Vision Transformer-based architectures.
- 2nd place at V-Medithon Intercollege Hackathon, Ranked 1 for Machine Learning model.
- 300+ problems solved on LeetCode with two consecutive 100 Days Badges (2025–2026). [↗](#)
- **Oracle Cloud Infrastructure Architecture: Data Science Professional**, Oracle University [↗](#)
- Patent accepted for publication on an IoT sensor-based smart biodegradable milk packaging system.